Answers learning task 1

2026 – 2027

In Questions 2.1, 2.8 and 2.17, provide a reasonable description of the sample space for each of the random experiments. There can be more than one acceptable interpretation of each experiment. Describe any assumptions you make.

Exercise 2.1: Each of three machined parts is classified as either above or below the target specification for the part.

Answer

$S = \{aaa, aab, aba, baa, abb, bab, bba, bbb\}$, where $a =$ above and $b =$ below

Exercise 2.8: The following two questions appear on an employee survey questionnaire. Each answer is chosen from the five-point scale 1 (never), 2, 3, 4, 5 (always).

- Is the corporation willing to listen to and fairly evaluate new ideas?
- How often are my coworkers important in my overall job performance?

Answer

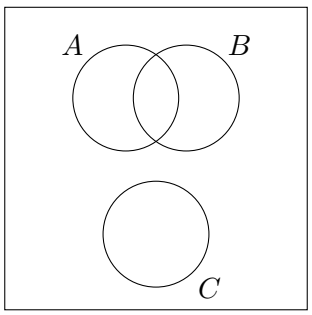
$S = \{(1,1), (1,2), \ldots, (1,5), (2,1), \ldots, (5,5)\}$; $n_S = 25$

Exercise 2.17: Calls are repeatedly placed to a busy phone line until a connection is achieved.

Answer

$S = \{c, bc, bbc, bbbc, \ldots\}$, where $b =$ busy and $c =$ connect

Exercise 2.20: Three events are shown on the Venn diagram in the following figure:

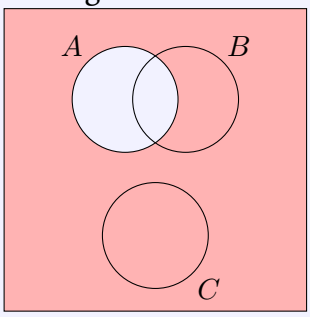


Reproduce the figure and shade the region that corresponds to each of the following events.

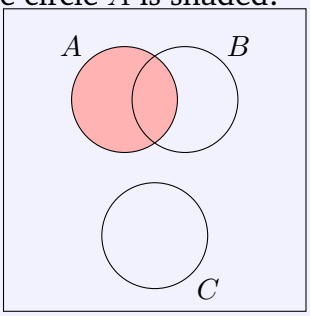
(a) A'
(b) $(A \cap B) \cup (A \cap B')$
(c) $(A \cap B) \cup C$
(d) $A \cap (B \cup C)'$
(e) $(A \cap B)' \cup C$

Answer

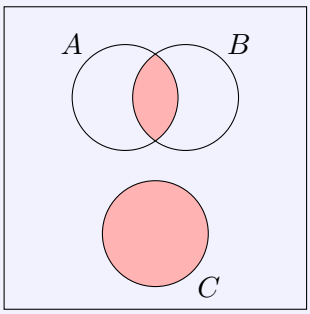
(a) A' is the region outside circle A:



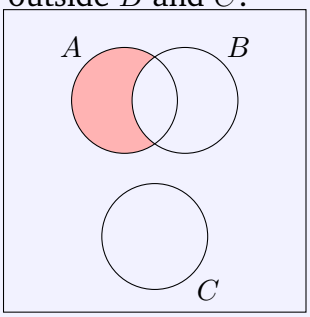
(b) $(A \cap B) \cup (A \cap B')$ simplifies to A, so the entire circle A is shaded:



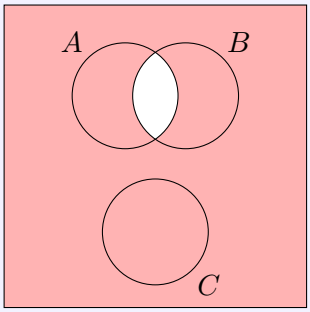
(c) $(A \cap B) \cup C$ is the intersection of A and B, plus the entire circle C:



(d) $A \cap (B \cup C)'$ is the portion of circle A that is outside B and C:



(e) $(A \cap B)' \cup C$ is everything outside the intersection of A and B, plus the entire circle C:



Exercise 2.29: The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. Define the events A and B as follows: $A = \{x \mid x < 65\}$ and $B = \{x \mid x > 45.5\}$.

Describe each of the following events:

(a) A'
(b) B'
(c) $A \cap B$
(d) $A \cup B$

Answer

(a) $\{x \mid x \geq 65\}$
(b) $\{x \mid 0 < x \leq 45.5\}$
(c) $\{x \mid 45.5 < x < 65\}$
(d) $\{x \mid 0 < x < \infty\}$

Exercise 2.31: A sample of two printed circuit boards is selected without replacement from a batch. Describe the (ordered) sample space for each of the following batches:

(a) The batch contains 90 boards that are not defective, seven boards with minor defects,

and three boards with major defects.

(b) The batch contains 90 boards that are not defective, eight boards with minor defects, and one board with major defects.

Answer

a. $S = \{gg, gm, gb, mg, mm, mb, bg, bm, bb\}$,
where $g =$ good, $m =$ minor and $b =$ bad

b. $S = \{gg, gm, gb, mg, mm, mb, bg, bm\}$

Exercise 2.55: The sample space of a random experiment is $\{a, b, c, d, e\}$ with probabilities $0.2, 0.1, 0.2, 0.3,$ and 0.2, respectively. Let A denote the event $\{a, b, c\}$, and let B denote the event $\{c, d, e\}$. Determine the following:

(a) $P(A)$ (b) $P(B)$ (c) $P(A')$
(d) $P(A \cup B)$ (e) $P(A \cap B)$

Answer

(a) 0.5 (b) 0.7 (c) 0.5
(d) 1 (e) 0.2

Exercise 2.74: If $P(A) = 0.4$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

(a) $P(A')$ (b) $P(A \cup B)$ (c) $P(A' \cap B)$
(d) $P(A \cap B')$ (e) $P[(A \cup B)']$ (f) $P(A' \cup B)$

Answer

(a) 0.6 (b) 0.5 (c) 0.1
(d) 0.3 (e) 0.5 (f) 0.7

Exercise 2.77: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch	High	70	10
Resistance	Low	15	5

(a) If a disk is selected at random, what is the probability that its scratch resistance is high and its shock resistance is high?

(b) If a disk is selected at random, what is the probability that its scratch resistance is high or its shock resistance is high?

(c) Consider the event that a disk has high scratch resistance and the event that a disk has high shock resistance. Are these two events mutually exclusive?

Answer

(a) 0.7

(b) 0.95

(c) No, because $P(A \cap B) = 0.70$. So $A \cap B \neq \emptyset$

Exercise 2.87: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch	High	70	10
Resistance	Low	15	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Determine the following probabilities:

(a) $P(A)$
(b) $P(B)$
(c) $P(A \mid B)$
(d) $P(B \mid A)$

Answer

(a) 0.85
(b) 0.8
(c) 0.875
(d) 0.8235

Exercise 2.91: The following table summarizes the analysis of samples of galvanized steel for coating weight and surface roughness:

		Coating Weight	
		High	Low
Surface	High	36	16
Roughness	Low	114	34

(a) If the coating weight of a sample is high, what is the probability that the surface roughness is high?

(b) If the surface roughness of a sample is high, what is the probability that the coating weight is high?

(c) If the surface roughness of a sample is low, what is the probability that the coating weight is low?

Answer

(a) 0.24

(b) 0.6923

(c) 0.2297

Exercise 2.106: Suppose that $P(A \mid B) = 0.25$, $P(A \mid B') = 0.35$, and $P(B) = 0.8$. What is $P(A)$?

Answer

0.27

Exercise 2.113: A batch of 30 injection-molded parts contains 6 parts that have suffered excessive shrinkage.

(a) If two parts are selected at random, and without replacement, what is the probability that the second part selected is one with excessive shrinkage?

(b) If three parts are selected at random, and without replacement, what is the probability that the third part selected is one with excessive shrinkage?

Answer

(a) 0.2

(b) 0.2